



Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU

Third Semester BE/B Tech Degree Examination: February/March 2026

MODEL QUESTION PAPER

Course: Object Oriented Programming with C++ and JAVA(BCSESC305A)

Duration: 3 Hours

Maximum Marks: 100

Instructions to the Students:

1. Answer any 5 full questions by choosing one full question from each module.
2. Missing data can be suitably assumed.

Q. No.	Question	Marks	CO	RL
Module 1				
1 a)	What is Object-Oriented Programming? Describe the key features of C++ as an object-oriented language.	6	1	L2
1 b)	How are classes and objects defined in C++? Illustrate with syntax and examples.	6	1	L2
1 c)	Write a C++ program to compute the area of Triangle, Square, Rectangle and Circle by overloading the area() function.	8	1	L3
OR				
2 a)	Describe Formatted and Unformatted input and output (I/O) functions available in C++ with syntax and suitable examples.	6	1	L2
2 b)	Demonstrate passing of objects as arguments to the function and returning in C++ with suitable examples.	6	1	L2
2 c)	Write a C++ program to demonstrate that a static member function can be called without creating an object.	8	1	L3
Module 2				
3 a)	What are templates in C++? Why are they used? How do templates help in achieving Generic programming?	6	2	L2
3 b)	Write a C++ program using vectors to insert elements, remove the elements, display size and content.	8	2	L2
3 c)	Give an example of using class templates with multiple parameters.	6	2	L3
OR				
4 a)	Explain different functions used in maps.	6	2	L2
4 b)	Write a C++ program for Bubble sort using a function template	8	2	L2
4 c)	Explain the difference between type and non-type template arguments.	6	2	L3
Module 3				



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Q. No.	Question	Marks	CO	RL
5 a)	Demonstrate the working of the for loop and for each loop with an example.	6	3	L2
5 b)	Define Array. Write a Java program to implement the addition of two matrices.	8	3	L2
5 c)	Explain the following operations with an example. (i) >>> (ii) Ternary operator	6	3	L2
OR				
6 a)	Define Classes and Objects with a suitable example	6	3	L2
6 b)	What are Constructors and Destructors? Explain two types of constructors with an example program.	8	3	L2
6 c)	Explain the use of this keyword in Java with an example.	6	3	L2
Module 4				
7 a)	Demonstrate the working of Super in Inheritance? Explain with an example.	8	4	L2
7 b)	Explain the concept of method overriding in java with an example.	6	4	L2
7 c)	Write a Java program to demonstrate multilevel inheritance.	6	4	L3
OR				
8 a)	Define Exception. Explain the Exception handling mechanism with an example.	8	4	L2
8 b)	Explain Multiple catch clauses in java.	6	4	L2
8 c)	Develop a Java program to raise a custom exception (user defined exception) for DivisionByZero using try, catch, throw and finally.	6	4	L3
Module 5				
9 a)	Explain the concept of importing packages in Java and provide an example demonstrating the usage of the import statement.	6	5	L2
9 b)	Define thread? Explain the different ways of creating threads.	8	5	L2
9 c)	Write a Java program to demonstrate producer consumer problem using synchronized threads.	6	5	L3
OR				



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Q. No.	Question	Marks	CO	RL
10 a)	Describe how interfaces facilitate multiple inheritance in Java.	6	5	L2
10 b)	Explain the life cycle of a thread with a neat diagram.	8	5	L2
10 c)	Write a Java program to illustrate the creation of threads using runnable class. (start method starts each of the newly created threads. Inside the run method, there is a sleep () for suspending the thread for 500 milliseconds.	6	5	L3

Q.P. Setter

Scrutinizer

BoE Chairman

CoE

Model Question Paper-I with effect from 2025-26 (CBCS Scheme)

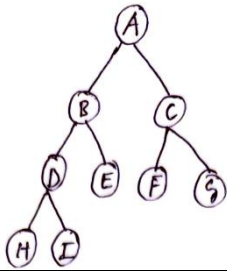
Third Semester B.E. Degree Examination Data Structures and Applications

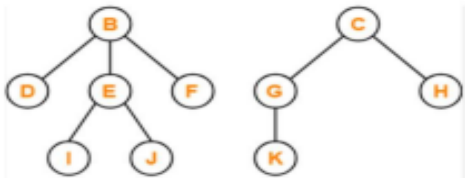
TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			CO	BL	Marks																																			
Q. 01	a	Define Data Structure. Classify data structures with one example each.	1	2	6																																			
	b	Develop a C program to read and display the details of n students. Each student record should include the following fields: Name, Roll Number, and Percentage.	1	3	7																																			
	c	Define a Sparse Matrix. Represent the given sparse matrix in triplet form. <table border="1"> <tr><td>15</td><td>0</td><td>0</td><td>22</td><td>0</td><td>-15</td></tr> <tr><td>0</td><td>11</td><td>3</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>-6</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>91</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>28</td><td>0</td><td>0</td><td>0</td></tr> </table>	15	0	0	22	0	-15	0	11	3	0	0	0	0	0	0	-6	0	0	0	0	0	0	0	0	91	0	0	0	0	0	0	0	28	0	0	0	1	3
15	0	0	22	0	-15																																			
0	11	3	0	0	0																																			
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0	0	0	0	0	0																																			
91	0	0	0	0	0																																			
0	0	28	0	0	0																																			
OR																																								
Q. 02	a	List and explain dynamic memory functions with syntax and examples.	1	2	6																																			
	b	Define pattern matching. Outline the Knuth–Morris–Pratt (KMP) algorithm and illustrate its working to find all occurrences of the following pattern in the given text. Pattern: ABCDABD Text: ABC ABCDAB ABCDABCDABDE	1	3	7																																			
	c	Construct a structure for representing a polynomial and write a C function to add two polynomials using arrays.	1	3	7																																			
Module-2																																								
Q. 03	a	A Custom calculator application takes a postfix expression as input. Design a function to evaluate that postfix expression using a stack as a data structure, and apply the same to evaluate the expression: $6\ 2\ /\ 3 - 4\ 2\ * +$	2	3	6																																			
	b	List the drawbacks of a linear queue. Explain how a circular queue overcomes these drawbacks. Write C functions to implement insert() and delete() operations for a circular queue.	2	3	7																																			
	c	Define a stack. Write C functions to implement push(), pop(), and display() operations for a stack using arrays.	2	3	7																																			
OR																																								
Q. 04	a	Convert the following infix expression to postfix using a stack. $A + B * (C \wedge D - E) \wedge (F + G * H) - I$	2	3	6																																			
	b	Explain the concept of Multiple Stacks. Illustrate your answer with a suitable diagram.	2	3	7																																			
	c	Develop a C program to implement the following operations on a Linear Queue using arrays: 1. Insertion of an element	2	3	7																																			

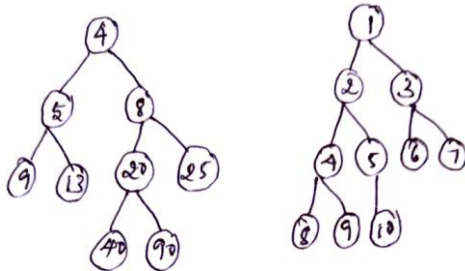
		2. Deletion of an element 3. Display of queue elements			
Module-3					
Q. 05	a	Define a linked list. Explain how a linked list is represented in C. Implement a stack using a singly linked list.	3	2	6
	b	Design a Library Management System using a dynamic linked list to store book details (Book ID, Title, and Author). Implement: addBook(), deleteBook(), displayBooks().	3	3	7
	c	Write a C function to add two polynomials using a linked list, and show the linked list representation of the two polynomials: $p(x)=3x^{14}+2x^8+1$ $q(x)=8x^{14}-3x^{10}+10x^6$	3	3	7
OR					
Q. 06	a	How to represent a Doubly Linked List (DLL) in C. Develop C functions to perform the following operations on integers. i. Insert a node at the beginning of DLL. ii. Delete a node at the end of the list.	3	2	6
	b	Implement the following list operations using a singly linked list. • reverseList() • Concatenate()	3	3	7
	c	Represent the given sparse matrix using a linked structure $\begin{matrix} 3 & 0 & 0 & 0 \\ 5 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 8 \\ 0 & 0 & 9 & 0 \end{matrix}$	3	3	7
Module-4					
Q. 07	a	Explain DFS and BFS with one example each.	4	2	6
	b	Construct an inorder threaded binary tree for the given binary tree and write its traversal function. 	4	3	7
	c	Define the following terms with suitable examples i. Degree of a Node ii. Level of a Binary Tree iii. Complete Binary Tree iv. Full Binary Tree	4	3	7
OR					
Q. 08	a	Define the following terminologies with examples. i. Graph ii. Complete graph iii. Multigraph iv. Directed & undirected graphs v. weighted and unweighted graphs	4	2	6

b	Define a Forest. Convert the given Forest into a Binary Tree. Perform inorder, preorder, and postorder traversal.	4	3	7
		4	3	7
c	Define Selection trees. Construct the Winner and Loser Tree for [10, 2, 7, 6, 5, 9, 12, 1]	4	3	7

Module-5

Q. 09	a What is chained hashing. Discuss its pros and cons. Construct the hash table to insert the keys: [7, 24, 18, 52, 36, 54, 11, 23] in a chained hash table of 9 memory locations. Use $h(k) = k \text{ mod } m$.	5	3	6
	b Define collision. List different methods to resolve collision. Apply Linear and Quadratic Probing for the keys [8, 3, 13, 6, 4, 10].	5	3	7
	c Define a Priority queue. Demonstrate C functions to implement the Max Priority queue with an example. i. Insert into the Max priority queue ii. Delete into the Max priority queue iii. Display Max priority queue	5	3	7

OR

Q. 10	a Define hashing. Explain different hashing functions with examples.	5	2	6
	b Define leftist tree. Give its declaration in C. Check whether the given binary tree is a leftist tree or not. Justify your answer. 	5	3	7
	c Apply Optimal Binary Search Tree (OBST) to derive an optimal solution for the following data. Keys: A B C D Probability: 0.1 0.2 0.4 0.3	5	3	7



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Third Semester BE/B Tech Degree Examination: February/March 2026

MODEL QUESTION PAPER

Course: Digital Design and Computer Organization (BCSPCC303)

Duration: 3 Hours

Maximum Marks: 100

Instructions to the Students:

1. Answer any 5 full questions by choosing one full question from each module.
2. Missing data can be suitably assumed.

Q. No.	Question	Marks	CO	RL
Module 1				
1 a)	State the following: i) Duality Principle ii) De-Morgan's theorem	6	1	2
1 b)	Simplify the following expression using K-map i) $f(A,B,C) = \sum m(0,1,3,5,6,7)$ ii) $f(A,B,C,D) = \sum m(0,3,5,7,8,9,10,12,13) + \sum d(1,6,11,14)$	8	1	3
1 c)	Obtain a minimum product of sums with a K-map: $F(w,x,y,z) = x'z' + wyz + w'y'z' + x'y$	8	1	3
OR				
2 a)	Using Quine-McClusky method determine a minimum sum-of-product for the function $f(w,x,y,z) = \sum m(0,1,2,3,5,6,7,11,13,15)$	8	1	3
2 b)	Explain different modelling styles used to write the HDL code in VERILOG with an example.	6	1	3
2 c)	Find the complement of the following function: $F(A,B,C,D) = \sum(0,1,2,3,4,8,9,12)$, convert to a NAND logic diagram that implements the given function.	6	1	3
Module 2				
3 a)	With a neat diagram, discuss the working of 4-bit adder-subtractor with overflow detection.	8	2	3
3 b)	Differentiate between latches and flip-flops.	4	2	2
3 c)	Implement the Verilog HDL for (i) Half adder (ii) Full adder using two half adders	8	2	3
OR				
4 a)	Implement a Full - Subtractor using Decoder.	6	2	3



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Q. No.	Question	Marks	CO	RL
4 b)	Design a combinational circuit that converts four-bit binary number to BCD. Implement the circuit using the basic logic gates.	8	2	3
4 c)	With timing diagram and truth table describe the operation of JK Flip-Flop. Write the HDL Verilog Code for JK Flip-Flop.	6	2	3
Module 3				
5 a)	Explain the basic operational concepts of the computer with a neat diagram.	6	3	2
5 b)	Discuss one address, two address, and three address instructions to carry out $C \leftarrow [A] + [B]$.	6	3	3
5 c)	Define addressing mode. Explain the different addressing modes with an example for each.	8	3	2
OR				
6 a)	Explain the following: a) Processor Clock b) Basic performance equation c) Clock rate d) Performance measurement	8	3	2
6 b)	Explain the following: (i) Byte addressability (ii) Big-endian assignment (iii) Little-endian assignment.	6	3	2
6 c)	Write an ALP to add a list of N numbers using indirect addressing mode.	6	3	3
Module 4				
7 a)	What is bus arbitration? Explain the different types of bus arbitration.	10	4	2
7b)	Define cache memory. Explain the different mapping functions used in cache memory.	10	4	2
OR				
8 a)	With neat diagrams, explain various methods for handling multiple interrupt requests from multiple devices.	10	4	2
8 b)	Draw a neat block diagram and explain the use of DMA controllers in a computer system.	10	4	2
Module 5				
9 a)	Explain the single-bus organization of computers and fundamental concepts with a neat diagram.	10	5	2



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Q. No.	Question	Marks	CO	RL
9 b)	Write and explain the control sequence for execution of the instruction Add (R3), R1	10	5	3
OR				
10 a)	Explain with an example the different types of hazards that occur during pipelining.	10	5	2
10 b)	Write and explain the control sequence for the execution of an unconditional branch instruction.	10	5	3

Q.P. Setter

Scrutinizer

BoE Chairman

CoE



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Third Semester BE/B Tech Degree Examination : February/March 2026

MODEL QUESTION PAPER

Course: Operating System (BCSPCC304)

Duration: 3 Hours

Maximum Marks: 100

Instructions to the Students:

1. Answer any 5 full questions by choosing one full question from each module.
2. Missing data can be suitably assumed.

Q. No.	Question	Marks	CO	RL																								
Module 1																												
1 a)	Define Operating System [OS]. Discuss the OS services with respect to Users and Designers perspective with a neat diagram.	7	CO1	L2																								
1 b)	Describe the role of operating systems across different computing environments.	7	CO1	L2																								
1 c)	What is a system program? Discuss various categories of system programs.	6	CO1	L2																								
OR																												
2 a)	Define Operating System [OS]. Demonstrate dual mode operations in OS with a neat schematic diagram.	7	CO1	L2																								
2 b)	Define system call? Categorize the different types of system calls. Explain three system calls in each category.	7	CO1	L2																								
2 c)	Describe the various structures of operating systems.	6	CO1	L2																								
Module 2																												
3 a)	What is a process? Illustrate different states of process with a neat diagram and also write the structure of the process control block.	5	CO2	L2																								
3 b)	Illustrate the producer-consumer problem using bounded buffer implementation with shared memory. Demonstrate with pseudo-code.	6	CO2	L2																								
3 c)	a) For the following set of processes:	9	CO2	L3																								
	<table><tr><th>Process</th><th></th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td></td><td>0</td><td>10</td></tr><tr><td>P2</td><td></td><td>2</td><td>1</td></tr><tr><td>P3</td><td></td><td>3</td><td>2</td></tr><tr><td>P4</td><td></td><td>4</td><td>1</td></tr><tr><td>P5</td><td></td><td>6</td><td>5</td></tr></table>				Process		Arrival Time	Burst Time	P1		0	10	P2		2	1	P3		3	2	P4		4	1	P5		6	5
	Process					Arrival Time	Burst Time																					
	P1					0	10																					
	P2					2	1																					
	P3					3	2																					
P4		4	1																									
P5		6	5																									



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Q. No.	Question	Marks	CO	RL																																																																						
	i. Draw the Gantt chart that illustrates the execution of these processes using FCFS, SJF (preemptive), Round Robin (quantum time = 3). ii. Find average turnaround around time and average waiting time for each of the scheduling algorithms.																																																																									
OR																																																																										
4 a)	What is IPC? What are the benefits of IPC? Describe different IPC models with suitable diagrams.	8	CO2	L2																																																																						
4 b)	Define context switching. Demonstrate its steps and interpret why it is considered an overhead.	6	CO2	L2																																																																						
4 c)	How do you differentiate thread from process? What are the advantages of threads? Explain different multithreaded models with necessary a diagram.	6	CO2	L2																																																																						
Module 3																																																																										
5 a)	Illustrate Peterson’s solution for the critical section problem.	6	CO3	L2																																																																						
5 b)	Define deadlock. What are the necessary conditions for deadlock to occur?	6	CO3	L2																																																																						
5 c)	Determine whether the following system is in safe state by using Bankers algorithm. <table><tr><th>Process</th><th colspan="3">Allocation</th><th colspan="3">Maximum</th><th colspan="3">Available</th></tr><tr><td></td><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P0</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P1</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P2</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P4</td><td>0</td><td>0</td><td>0</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> If a request for process p1 arrives for (1, 0, 2) can the request be granted immediately?	Process	Allocation			Maximum			Available				A	B	C	A	B	C	A	B	C	P0	0	1	0	7	5	3	3	3	2	P1	2	0	0	3	2	2				P2	3	0	2	9	0	2				P3	2	1	1	2	2	2				P4	0	0	0	4	3	3				8	CO3	L3
Process	Allocation			Maximum			Available																																																																			
	A	B	C	A	B	C	A	B	C																																																																	
P0	0	1	0	7	5	3	3	3	2																																																																	
P1	2	0	0	3	2	2																																																																				
P2	3	0	2	9	0	2																																																																				
P3	2	1	1	2	2	2																																																																				
P4	0	0	0	4	3	3																																																																				



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Q. No.	Question	Marks	CO	RL
OR				
6 a)	What is a resource allocation graph? Consider an example to explain how it is very useful in describing a deadly embrace.	6	CO3	L2
6 b)	Explain different methods to recover from deadlocks.	6	CO3	L2
6 c)	What is a semaphore? State a Dining Philosopher problem gives a solution using semaphore.	8	CO3	L2
Module 4				
7 a)	Illustrate contiguous memory allocation with neat diagrams	5	CO4	L2
7 b)	Illustrate demand paging with a diagram showing page fault handling.	5	CO4	L2
7 c)	Consider the page reference string : 1,0,7,1,0,2,1,2,3,0,3,2,4,0,3,6,2,1 for a memory with 3 frames. Determine the number of page faults using FIFO,optimal and LRU replacement algorithms. Which algorithm is most efficient?	10	CO4	L3
OR				
8 a)	Describe the structure of a page table with an example.	5	CO4	L2
8 b)	Illustrate paging hardware with TLB.	5	CO4	L2
8 c)	A process makes page references 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Compute the number of page faults using (a) FIFO, (b) LRU, (c) Optimal algorithms for 3 frames.	10	CO4	L2
Module 5				
9 a)	Define a file in the context of computer systems. Explain the key attributes of a file and describe the various file operations.	6	CO5	L1
9 b)	Illustrate the different file allocation methods	6	CO5	L2



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Q. No.	Question	Marks	CO	RL
9 c)	Describe the access matrix approach to system protection, treating domains as objects, and discuss its implementation methods.	8	CO5	L3
OR				
10 a)	Explain protection in operating systems along with types of file access and access control mechanisms.	6	CO5	L2
10 b)	Discuss various directory structures with neat diagrams.	6	CO5	L2
10 c)	A Drive as 200 Cylinders 0 to 129 Head starts at 53 to serve the request queue 98,183,37,122,14,124,65,67. Draw disk head schedule Diagram using FCFS, SCAN and C-SCAN Algorithms.	8	CO5	L3

Q.P. Setter

Scrutinizer

BoE Chairman

CoE